

OIS-008 — Knowledge Engine

Status: Foundational

Authority: Implementation

Depends on:

- OIS-000 through OIS-007
 - OAS-014 — Knowledge Architecture
 - OCS-000 through OCS-032
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Purpose

This specification defines the implementation responsibilities of the Knowledge Engine.

The Constitution defines the nature of Knowledge.

The Architecture defines the organization of Knowledge.

This specification defines the executable engine responsible for constructing, maintaining, querying, and evolving constitutional knowledge.

The Knowledge Engine SHALL execute constitutional knowledge behavior.

It SHALL never redefine constitutional meaning.

1. Implementation Objective

The objective of the Knowledge Engine is to transform constitutional relationships into an operational environment capable of supporting continuous understanding.

The Engine SHALL preserve:

Semantic relationships.

Historical continuity.

Knowledge evolution.

Architectural integrity.

Explainability.

Reconstructability.

Knowledge SHALL remain constitutionally authoritative.

2. Knowledge Construction

The Engine SHALL construct operational knowledge from:

Constitutional identities.

Events.

Relationships.

Schemas.

Libraries.

Conversations.

Reasoning.

No operational shortcut SHALL bypass constitutional history.

3. Knowledge Graph

Implementations MAY represent knowledge using:

Property graphs.

Semantic graphs.

Relational projections.

Document relationships.

Vector-assisted structures.

Hybrid representations.

Representation SHALL remain implementation-specific.

Constitutional relationships SHALL remain invariant.

4. Semantic Resolution

The Engine SHALL resolve semantic relationships among constitutional constructs.

Resolution SHALL preserve:

Meaning.

Context.

Historical lineage.

Architectural compatibility.

Equivalent semantic understanding SHALL remain observable.

5. Knowledge Queries

The Engine SHALL support constitutional queries including:

Identity traversal.

Relationship exploration.

Historical reconstruction.

Reasoning exploration.

Conversation discovery.

Evidence lineage.

Learning history.

Query execution SHALL preserve constitutional semantics.

6. Knowledge Evolution

Knowledge SHALL evolve through constitutional Events.

Implementation SHALL preserve:

Previous understanding.

Current understanding.

Relationship evolution.

Historical continuity.

Operational optimization SHALL never overwrite constitutional history.

7. Knowledge Projections

The Engine MAY generate projections including:

Semantic indexes.

Relationship summaries.

Search structures.

Recommendation structures.

Exploration maps.

Every projection SHALL remain derivable from constitutional history.

8. Knowledge Federation

Knowledge Engines MAY federate.

Federation SHALL preserve:

Local ownership.

Canonical identity.

Relationship integrity.

Historical attribution.

Semantic equivalence.

Federation SHALL remain implementation-independent.

9. Recovery

Recovery SHALL reconstruct constitutional knowledge using:

Identity.

Schemas.

Events.

Relationships.

Architectural specifications.

Recovery SHALL preserve semantic continuity.

10. Verification

Verification SHALL demonstrate:

Semantic preservation.

Relationship correctness.

Historical reconstruction.

Equivalent query behavior.

Federation compatibility.

Equivalent constitutional knowledge behavior.

11. Knowledge Pattern

The Knowledge Engine SHALL optimize discovery without sacrificing explainability.

Search SHALL remain subordinate to understanding.

Performance SHALL remain subordinate to constitutional integrity.

12. Knowledge Engine Test

A Knowledge Engine satisfies this specification when independent implementations produce constitutionally equivalent knowledge exploration, historical reconstruction, semantic relationships, and reasoning discovery despite differing internal representations and execution strategies.

Conformance

An implementation conforms to this specification when constitutional knowledge remains operationally executable, semantically equivalent, historically continuous, and architecturally reconstructable independently of implementation technology.

Implementation Principle

Traditional knowledge systems retrieve information.

The Knowledge Engine cultivates understanding.

Its purpose is not merely to answer questions,

but to preserve the constitutional landscape in which every answer remains connected to its evidence,

its reasoning,

its history,

and the Questions that first made it necessary.

End of Specification OIS-008